

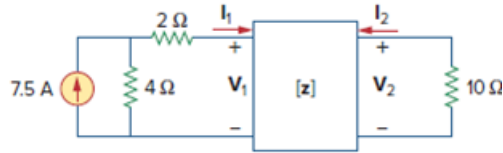
Problem

For the circuit shown in Fig. 19.73, let

$$[z] = \begin{bmatrix} 10 & -6 \\ -4 & 12 \end{bmatrix} \Omega$$

Find I_1 , I_2 , V_1 , and V_2 .

Figure 19.73



Step-by-step solution

Step 1 of 7

Refer to Figure 19.73 in the text book.

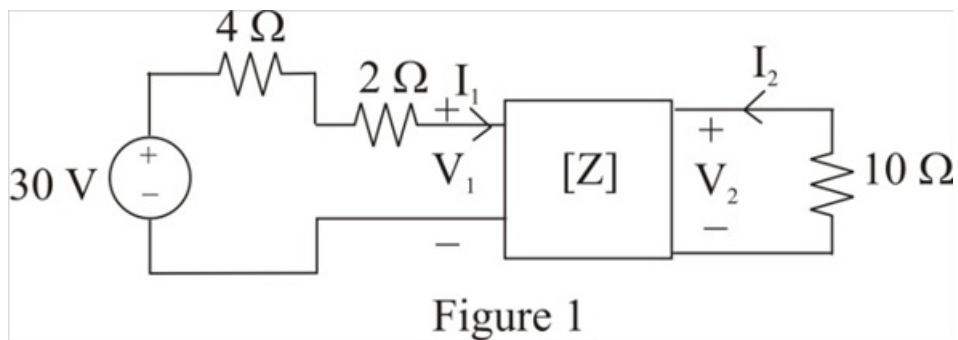
Convert the current source to voltage source. Calculate the value of the voltage source.

$$\begin{aligned} V &= (7.5)(4) \\ &= 30 \text{ V} \end{aligned}$$

Therefore, the value of the voltage source is 30 V.

Step 2 of 7

Consider the circuit diagram shown in Figure 1.



Step 3 of 7

Consider the expression for the voltage V_2 in terms of z-parameters.

$$V_2 = z_{21}I_1 + z_{22}I_2$$

Substitute -4Ω for z_{21} , 12Ω for z_{22} and $-I_2(10)$ for V_2 .

$$-I_2(10) = (-4)I_1 + (12)I_2$$

$$(-22)I_2 = (-4)I_1$$

$$I_2 = \left(\frac{4}{22}\right)I_1$$

$$I_2 = \left(\frac{2}{11}\right)I_1 \dots\dots (1)$$

Therefore, the expression for the current I_2 is $\left(\frac{2}{11}\right)I_1$.

Step 4 of 7

Consider the expression for the voltage V_1 in terms of the source voltage.

$$V_1 = 30 - I_1(2 + 4)$$

$$V_1 = 30 - I_1(6) \dots\dots (2)$$

Consider the expression for the voltage V_1 in terms of z-parameters.

$$V_1 = z_{11}I_1 + z_{12}I_2$$

Substitute $\left(\frac{2}{11}\right)I_1$ for I_2 , $30 - I_1(6)$ for V_1 , -6Ω for z_{12} and 10Ω for z_{11} .

$$30 - I_1(6) = (10)I_1 + (-6)\left(\frac{2}{11}\right)I_1$$

$$30 - I_1(6) = (10)I_1 - \left(\frac{12}{11}\right)I_1$$

$$30 - I_1(6) = (10)I_1 - (1.09)I_1$$

$$30 = (10)I_1 + (6)I_1 - (1.09)I_1$$

$$30 = (14.91)I_1$$

$$I_1 = \frac{30}{14.91}$$

$$I_1 = 2.01 \text{ A}$$

Therefore, the value of the current I_1 is 2.01 A.

Step 5 of 7

Recall equation (1).

$$I_2 = \left(\frac{2}{11}\right)I_1$$

Substitute 2.01 A for I_1 .

$$\begin{aligned} I_2 &= \left(\frac{2}{11}\right)(2.01) \\ &= 365.46 \text{ mA} \end{aligned}$$

Therefore, the value of the current I_2 is 365.46 mA.

Step 6 of 7

Recall equation 2.

$$V_1 = 30 - I_1(6)$$

Substitute 2.01 A for I_1 .

$$\begin{aligned} V_1 &= 30 - (2.01)(6) \\ &= 30 - 12.06 \\ &= 17.94 \text{ V} \end{aligned}$$

Therefore, the value of the voltage V_1 is 17.94 V.

Step 7 of 7

Consider the expression for the voltage V_2 .

$$V_2 = -I_2(10)$$

Substitute 365.46 mA for I_2 .

$$\begin{aligned} V_2 &= -(365.46 \text{ m})(10) \\ &= -3.655 \text{ V} \end{aligned}$$

Therefore, the value of the voltage V_2 is -3.655 V.